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Wettability Quantification in Mixed-Wet Rocks Using a New NMR-Based Method

Saurabh Tandon, Chelsea Newgord, and Zoya Heidari, The University of Texas at Austin

Summary

Nuclear magnetic resonance (NMR) measurements have been attractive options for wettability characterization of reservoir rocks as they are sensitive to the type of fluid in contact with the grain surface. Several NMR-based wettability indices are documented in previous publications. Most of these methods require extensive calibration or involve complex inversion algorithms, which makes them computationally expensive and complicates their applicability in mixed-wet multimodal rocks. In this paper, we introduce a new NMR-based wettability index and verify its reliability in pore-scale and core-scale domains using numerical simulations and experimental measurements, respectively. This new index requires calibration at fully water-saturated water-wet and fully hydrocarbon-saturated hydrocarbon-wet states and can be applied to mixed-wet rocks at any fluid saturation level.

This new NMR-based wettability index is a function of the transverse magnetization (T_2) of mixed-wet rocks, the bulk relaxivity and saturation of each fluid, and the T_2 distributions for fully water-wet and hydrocarbon-wet samples of the same rock type. The reliability of the new index was first tested in the pore-scale domain. For this part, we selected several pore-scale microcomputed tomography (CT) images of carbonate and sandstone rocks. We used a previously developed finite volume simulator to model the T_2 responses in these images at fully water-wet and fully hydrocarbon-wet wettability states. Then we generated synthetic partially saturated mixed-wet samples and simulated T_2 responses in these synthetic images. We used the simulated T_2 results for determining their NMR-based wettability index and verified its applicability in the pore-scale domain.

Next, we tested the reliability of the new NMR-based wettability index in the core-scale domain using NMR measurements in four Texas Cream (TC) rock samples, obtained from the Edwards formation. We altered the wettability of the cores to be water-, hydrocarbon-, and mixed-wet by injecting brine, a naphthenic acid in decane solution, or an anionic surfactant solution. We quantified the wettability of these samples using the Amott-Harvey (AH) index and contact angle measurements. Next, we measured the T_2 distribution of these samples at different fluid saturation levels. Finally, we quantified the wettability values of these core samples using the new NMR-based index and compared them to those obtained from the AH index and contact angle measurements. We documented successful verification of the proposed method on samples with wettability ranging from -0.90 to 0.98 and from -0.6 to 0.5 (independently quantified using the AH method) in the pore- and core-scale domains, respectively. Results demonstrated that the new NMR-based wettability index reliably estimates the wettability of mixed-wet rocks in a wide range of wettability states. The new wettability index can potentially improve the speed and reliability of NMR-based wettability characterization and is promising for log-scale wettability assessment in mixed-wet rocks.

Introduction

Fluid molecular interactions at the grain/fluid interface play an important role in establishing mobility of different fluid phases and determining the ultimate hydrocarbon recovery of the reservoir. Rock wettability is the grain property that characterizes the ability of fluids to adhere to grain surfaces in the presence of another fluid phase. It can also be defined as the ability of a fluid to adhere to a solid surface when another immiscible phase is present. This property captures the effect of surface tension and interfacial interactions between the grain and fluid phases (Anderson 1986a, 1986b).

Most of the sandstones found in oil and gas reservoirs are water-wet but the carbonate reservoirs could become hydrocarbon- or mixed-wet. The most common cause of wettability alteration in such reservoir rocks is a result of precipitation of heavy hydrocarbon components, such as asphaltenes, on the grain surface (Buckley et al. 1996, 1997; Yu and Buckley 1997; Buckley 1998). In addition, the rock wettability can be modified using large-scale surfactant treatments during enhanced oil treatments to increase hydrocarbon recovery. For reservoir rocks, wettability is important to quantify as it impacts rock relative permeability, flow rates from production wells, overall hydrocarbon recovery, and choice of enhanced oil recovery or improved oil recovery methods (Donaldson et al. 1969; Anderson 1986a, 1986b).

Rock wettability for reservoir rocks can be experimentally quantified in the laboratory using the AH and the United States Bureau of Mines (USBM) indices (Amott 1959; Donaldson et al. 1969). The AH index measures the change in saturation of water and hydrocarbon phases during spontaneous and forced drainage and imbibition cycles. The AH index is determined by placing the core sample in a sealed Amott cell and carefully measuring the volume of water or hydrocarbon produced from the core sample (Amott 1959). The AH index (I_{AH}) is given by

$$I_{AH} = \frac{S_{WSP} - S_{WIRR}}{(1 - S_{OR}) - S_{WIRR}} - \frac{(1 - S_{OR}) - S_{WSP}}{(1 - S_{OR}) - S_{WIRR}} \quad (1)$$

In Eq. 1 S_{OR} is the residual oil saturation, S_{WIRR} is the irreducible water saturation, S_{WSP} is the water saturation at the end of spontaneous imbibition of brine, and S_{WSP0} is the water saturation after spontaneous imbibition of hydrocarbon phase.

The USBM index (I_{USBM}) is based on quantifying the work done by one fluid to displace another fluid in a core sample (Donaldson et al. 1969). Measuring the USBM index usually employs a centrifuge in which water and hydrocarbon saturations and capillary pressures are determined during forced drainage and imbibition cycles. The USBM index is given by

$$I_{USBM} = \log \frac{A_I}{A_D}, \quad (2)$$

where A_I and A_D are the areas under the capillary pressure curves at the end of imbibition and drainage cycles. Although the AH and the USBM indices provide the reliable estimates of wettability for core samples, these methods have their limitations (Anderson 1986b; Buckley et al. 1996; Ma et al. 1999). Sample preparation, preservation, storage, and test conditions such as laboratory temperature and humidity are critical in measuring wettability because wettability is sensitive to temperature as well as oxidation of crude hydrocarbons (Ma et al. 1999; Al-Muthana et al. 2012). In addition, both measurements can also be slow and time-consuming (Buckley et al. 1996; Al-Muthana et al. 2012). Furthermore, the AH and the USBM methods provide only laboratory measurements of rock wettability and are unable to estimate wettability directly from field data.

NMR measurements provide an indirect method for quantifying rock wettability as they are sensitive to the volume of fluid in the core samples as well as the type of fluid in contact with core grain surfaces (Brown and Fatt 1956; Borgia et al. 1991; Howard 1998). Several publications document using NMR experimental measurements, numerical simulations, and complex inversion algorithms for quantifying wettability of rock samples. The sensitivity of NMR measurements to wettability was first documented in the publication of Brown and Fatt (1956). They measured the relaxation rate of different fluids in packs of water- and hydrocarbon-wet beads and showed that relaxation rates vary linearly with the fraction of hydrocarbon-wet beads. In addition, Borgia et al. (1991), Hsu et al. (1992), and Howard (1998) showed that NMR T_2 measurements are sensitive to the type of fluid in contact with rock grains. Additional work by Guan et al. (2002) documented a linear relationship between the arithmetic mean of relaxation times and the AH index of mixed-wet cores at the residual oil saturation and the irreducible water saturation. Johannesen et al. (2007) analyzed the NMR T_2 responses of carbonate samples and demonstrated that the dominant T_2 peak can be directly correlated to rock wettability. Valori et al. (2017) quantitatively assessed the rock wettability in carbonate samples using the three-dimensional (3D), Diffusivity (D)–Longitudinal (T_1)–Transverse (T_2) magnetization, NMR responses to separate water and hydrocarbon signals from the magnetization decay.

NMR-based wettability indices provide a quantitative method to estimate rock wettability from NMR measurements. Al-Mahrooqi et al. (2006) proposed an NMR-based wettability index derived on the log-mean of the T_2 distributions at irreducible water and residual oil saturations. The Al-Mahrooqi-Grattoni-Muggeridge (AGM) wettability index (I_{AGM}) is given by

$$I_{AGM} = \frac{T_{2LM}^{WC} - T_{2LM}^{OR}}{T_{2LM}^{OR}}, \quad \dots \quad (3)$$

where T_{2LM}^{WC} and T_{2LM}^{OR} are the log-mean of the T_2 distributions at irreducible water and residual oil saturations, respectively. Fleury and Deflandre (2003) defined the pore-scale rock wettability as the difference in fraction of surface areas of the grains in contact with different fluids. They defined the pore-scale wettability (I_R) of rock samples as

$$I_R = f_W - f_{HC}, \quad \dots \quad (4)$$

where f_W is the fraction of surface area of the rock in contact with water, and f_{HC} is the fraction of surface area of the rock in contact with hydrocarbon. Fleury-Deflandre (FD) defined the NMR-based wettability index (I_{FD}) via

$$I_{FD} = \frac{S_W \left(\frac{1}{T_{2WP}} - \frac{1}{T_{BW}} \right) - C_{WH} (1 - S_W) \left(\frac{1}{T_{2HCP}} - \frac{1}{T_{BHC}} \right)}{S_W \left(\frac{1}{T_{2WP}} - \frac{1}{T_{BW}} \right) + C_{WH} (1 - S_W) \left(\frac{1}{T_{2HCP}} - \frac{1}{T_{BHC}} \right)}, \quad \dots \quad (5)$$

where S_W is the water saturation, T_{2WP} is the dominant peak of the water phase in the T_2 distribution at irreducible water saturation, and T_{2HCP} is the dominant peak of the hydrocarbon phase T_2 distribution at residual oil saturation. T_{BW} and T_{BHC} are the bulk relaxation constants of water and hydrocarbon phases, respectively. C_{WH} is the ratio of water to hydrocarbon transverse surface relaxivity which can be calculated as

$$C_{WH} = \frac{\left(\frac{1}{T_{2WP}} - \frac{1}{T_{BW}} \right)_{WW}}{\left(\frac{1}{T_{2HCP}} - \frac{1}{T_{BHC}} \right)_{OW}}. \quad \dots \quad (6)$$

The subscripts WW and OW in Eq. 6 indicate the T_2 distributions of water-saturated strongly water-wet and hydrocarbon-saturated strongly hydrocarbon-wet rock samples of the same rock type, respectively. Fleury and Deflandre (2003) recommended using dominant NMR T_2 peak at residual oil saturation and irreducible water saturation as inputs for calculating the NMR-based wettability of the rock sample.

Chen et al. (2006) presented an NMR-based wettability index derived from the ratios of effective hydrocarbon and water surface relaxivities of the mixed-wet sample at residual hydrocarbon and irreducible water saturations, respectively. Chen et al. (2006) defined the NMR-based wettability index as

$$I_{CHF} = \frac{S_W \left(\frac{1}{T_{2WP}} - \frac{1}{T_{BW}} \right)}{\left(\frac{1}{T_{2WP}} - \frac{1}{T_{BW}} \right)_{WW}} - \frac{(1 - S_W) \left(\frac{1}{T_{2HCP}} - \frac{1}{T_{BHC}} \right)}{\left(\frac{1}{T_{2HCP}} - \frac{1}{T_{BHC}} \right)_{OW}}, \quad \dots \quad (7)$$

where I_{CHF} is the Chen-Hirasaki-Flaum (CHF) NMR-based wettability index. I_{CHF} is calculated using either the dominant peak or the log-mean of the T_2 distribution.

Looyestijn and Hofman (LH) (2006) and Looyestijn (2008) proposed a fluid substitution algorithm (LH algorithm) that is based on calibrating the NMR response of the partially saturated mixed-wet rock sample with that of the fully water-saturated water-wet rock samples and hydrocarbon-saturated hydrocarbon-wet. The inversion proposed by the LH algorithm also outputs fluid saturation and can be adapted for fluids with a distribution of T_{BHC} values. The LH algorithm has been adapted for different fluid and pore wettability functions in the publication of Al-Muthana et al. (2012).

On the basis of the brief review of the aforementioned NMR-based wettability indices, we summarize some of their advantages and limitations in **Table 1**. The wettability indices documented in Table 1 are mostly used in conventional reservoirs such as sandstones and carbonates. In this paper, we introduce a new NMR-based wettability index to quantify the wettability in formations with complex pore structures for partially water- and hydrocarbon-saturated conditions. The new NMR-based wettability index is based on comparing the T_2 distributions of fully water and fully hydrocarbon saturated samples with that of partially saturated mixed-wet rock sample of the same rock type. The introduced wettability index enables wettability determination at any fluid saturation state in mixed-wet rocks. In addition, it can be applied to rocks with multimodal pore-size distribution. We verify the reliability of the newly introduced wettability index in the pore-scale domain using pore-scale numerical simulations of magnetization decay. We then verify the reliability of the newly introduced wettability index in the core-scale domain using core-scale NMR measurements.

Index	AGM	FD	CHF	LH
Advantages	Could be easily applied to any reservoir rock sample. Does not require calibration curves or prior knowledge of fluid relaxivities and saturations.	Tested for a wide range of USBM wettability states from -0.3 to 0.9.	Can quantify the sensitivity of NMR measurements to changes in surface relaxivity and the fractional fluid surface areas. Tested for a range of AH wettability states from -0.6 to 0.9.	Can be used for rocks with multimodal pore-size distributions. Provides both fluid saturations and rock wettability. Can be applied to any partially saturated mixed-wet rock sample. Tested with both the core and log-scale NMR measurements.
Limitations	Can be calculated only if NMR responses are available at irreducible water saturation and residual oil saturation. Has a limited physical basis as the dominant T_2 values could vary due to changes in pore-structure and surface properties of the rock samples.	Derived for rocks with unimodal pore-size distributions. Applied only to rock samples at irreducible water and residual oil saturation states. Requires knowledge of bulk fluid relaxivities and fluid saturations. Requires calibration in a water-wet rock and a hydrocarbon-wet rock. Could not successfully predict the wettability of carbonate core samples.	Tested only on Berea Sandstone samples. Derived for rocks with unimodal pore-size distributions. Applied only to rock samples at irreducible water and residual oil saturation states. Requires knowledge of bulk fluid relaxivities and fluid saturations. Requires calibration in a water-wet rock and a hydrocarbon-wet rock.	Requires knowledge of bulk fluid relaxivities and fluid saturations. Requires calibration in a water-wet rock and assumes a constant ratio for water and hydrocarbon surface relaxivities. Requires separate functions for describing pore wettability and water saturation. Could be time-consuming. Might provide nonunique results for the different fluid and pore-size distributions.

Table 1—Advantages and limitations of different NMR-based wettability indices.

Method

In this section, we describe the derivation of the new NMR-based wettability index as well as the algorithms for its pore-scale simulation-based and core-scale experimental verification. The inputs to the new NMR-based wettability index are the T_2 distributions and fluid saturations of the partially saturated mixed-wet rock samples. In the pore-scale domain, the water saturation was estimated by analysis of pore-scale images of partially saturated mixed-wet rock samples. In the core-scale domain the water saturation was obtained gravimetrically before performing NMR T_2 measurements. The new index also requires T_2 distributions of the completely water-saturated water-wet and completely hydrocarbon-saturated hydrocarbon-wet samples of the same rock type. The index outputs the wettability of the mixed-wet rock at any level of fluid saturation.

Introduction of the New NMR-Based Wettability Index. The new NMR-based wettability index is based on quantifying the difference in fractional surface areas covered by different fluids in a partially saturated mixed-wet rock sample using NMR measurements. Here we present a new NMR-based wettability index in terms of the T_2 distribution. A similar index could be derived using T_1 measurements. The new NMR-based wettability index (I_{NMR}) can be written using the harmonic mean of T_2 distribution of the partially saturated mixed-wet rock sample ($T_{2\text{HWM}}$) and the harmonic mean of bulk fluid relaxivities of saturating fluids (T_{BHM}) as

$$I_{\text{NMR}} = \frac{2 \left(\frac{1}{T_{2\text{HWM}}} - \frac{1}{T_{\text{BHM}}} \right) - \left[\left(\frac{1}{T_{2\text{WHM}}} - \frac{1}{T_{\text{BW}}} \right) + \left(\frac{1}{T_{2\text{HCHM}}} - \frac{1}{T_{\text{BHC}}} \right) \right]}{\left(\frac{1}{T_{2\text{WHM}}} - \frac{1}{T_{\text{BW}}} \right) - \left(\frac{1}{T_{2\text{HCHM}}} - \frac{1}{T_{\text{BHC}}} \right)}, \quad \dots \quad (8)$$

where $T_{2\text{WHM}}$ is the harmonic mean of the T_2 distribution of a water-saturated water-wet rock and $T_{2\text{HCHM}}$ is the harmonic mean of the T_2 distribution of a hydrocarbon-saturated hydrocarbon-wet rock. Appendix A describes the detailed derivation of Eq. 8 from the T_2

response of a partially saturated mixed-wet rock sample. The value of T_{2HM} can be calculated from the T_2 distribution of the partially saturated mixed-wet rock sample after normalizing it to rock porosity as

$$\frac{1}{T_{2HM}} = \sum_{i=0}^{\infty} \phi_i(r) \frac{1}{T_{2i}}, \quad \dots \dots \dots (9)$$

where ϕ_i and T_{2i} are the normalized pore-volume fraction and the decay time in pore i . Similar calculations should be performed using NMR T_2 responses of water-saturated water-wet and hydrocarbon-saturated hydrocarbon-wet samples for obtaining T_{2WHM} and T_{2HCHM} , respectively. These model parameters must be obtained for the same rock type as the partially saturated mixed-wet sample. The value of T_{BHM} can be calculated using water saturation and bulk relaxivities of water and hydrocarbon phases as

$$\frac{1}{T_{BHM}} = \frac{S_W}{T_{BW}} + \frac{1 - S_W}{T_{BHC}}. \quad \dots \dots \dots (10)$$

Pore-Scale Numerical Verification of the New NMR-Based Wettability Index. In this section, we first describe the method adopted for processing the micro-CT scan images and generating synthetic samples with different wettability states and different water/hydrocarbon saturations. Then, we describe the method for simulating magnetization decay for different samples with different saturation and wettability states. We use the magnetization decay signals to obtain the T_2 distributions that are then used to verify the applicability of the new NMR-based wettability index (Eq. 8) in the pore-scale domain.

Processing Micro-CT Scan Images. Fig. 1 shows the carbonate and sandstone samples used for numerical simulations in this paper. Table 2 lists properties of rock samples (subsections of the samples shown in Fig. 1) used in NMR simulations. Samples S1 and S2 are sandstone samples, scanned and digitized by the Petroleum Engineering and Rock Mechanics group, Imperial College London (Dong 2007). Samples C3 through C9 are taken from the Estailades Carbonate scanned and digitized by the Petroleum Engineering and Rock Mechanics group Imperial College London (Dong 2007). Samples C10 through C15 are taken from the Scurry Area Canyon Reef Operators Committee (SACROC) Unit, located in the Midland Basin, West Texas and scanned and digitized at Texas A&M University, College Station (Oyewole et al. 2016).

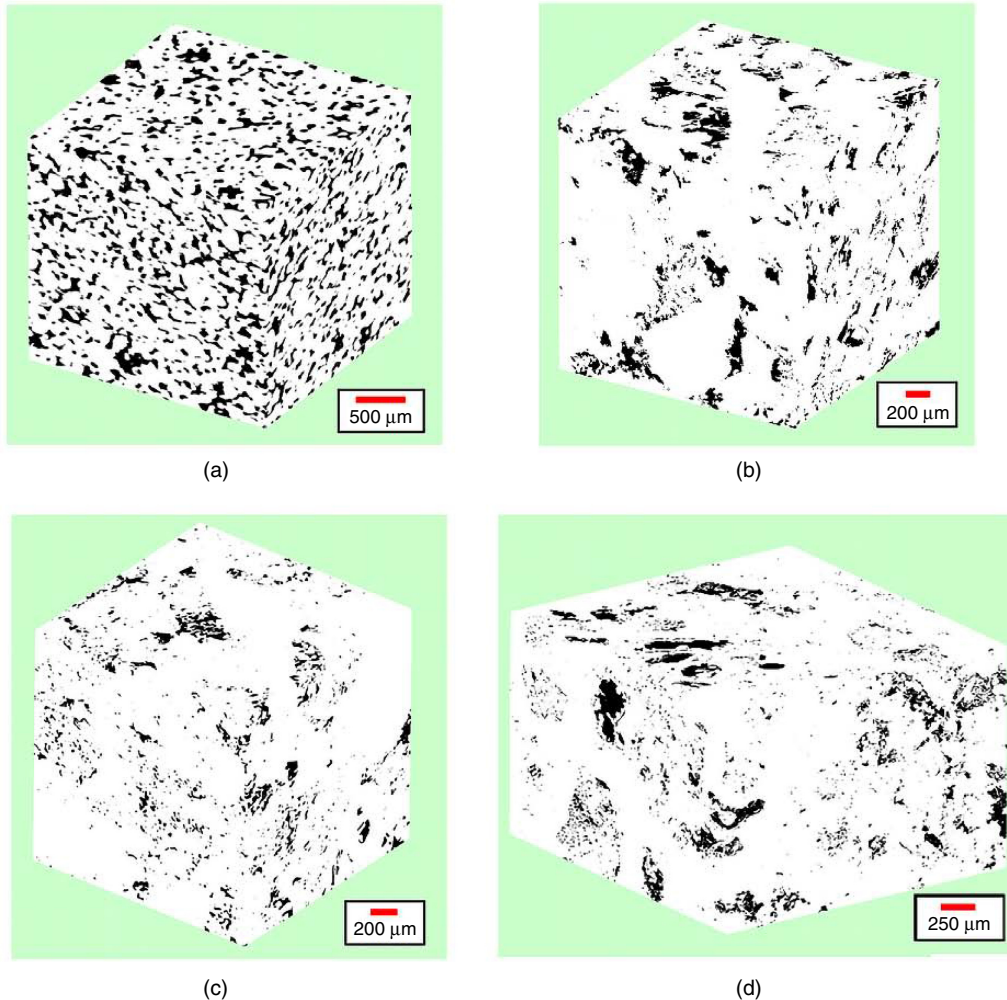


Fig. 1—Pore-scale images of (a) one sandstone and (b, c, and d) three carbonate rock types used as inputs for numerical simulations.

Sample Name	Resolution ($\mu\text{m}/\text{pixel}$)	Size (pixel^3)	Sample Porosity (ϕ_T)	Water Saturation (S_W)	Fraction of Grain Surface Covered by Water (f_W)	Pore-Scale Wettability (I_R)
S1	8.96	$200 \times 200 \times 200$	0.16	0.97	0.95	0.90
S2	8.96	$200 \times 200 \times 200$	0.17	0.03	0.05	-0.90
C3	2.64	$200 \times 200 \times 200$	0.10	0.91	0.87	0.74
C4	2.64	$200 \times 200 \times 200$	0.04	0.16	0.22	-0.56
C5	2.64	$200 \times 200 \times 200$	0.05	0.25	0.34	-0.32
C6	2.64	$200 \times 200 \times 200$	0.03	0.47	0.69	0.38
C7	2.64	$200 \times 200 \times 200$	0.06	0.84	0.81	0.62
C8	2.64	$200 \times 200 \times 200$	0.15	0.99	0.99	0.98
C9	2.64	$200 \times 200 \times 200$	0.13	0.05	0.11	-0.78
C10	2.00	$200 \times 200 \times 200$	0.03	0.69	0.74	0.48
C11	2.00	$200 \times 200 \times 200$	0.05	0.66	0.66	0.32
C12	2.00	$200 \times 200 \times 200$	0.02	0.37	0.52	0.04
C13	2.00	$200 \times 200 \times 200$	0.06	0.36	0.46	-0.08
C14	2.00	$200 \times 200 \times 200$	0.08	0.20	0.34	-0.32
C15	2.00	$200 \times 200 \times 200$	0.03	0.11	0.25	-0.50

Table 2—Image properties of the rock samples used as inputs to numerical simulations.

Development of Synthetic Pore-Scale Partially Saturated Mixed-Wet Samples. We tested the applicability of the newly introduced wettability index through numerical modeling by developing synthetic cases to cover a wide range of synthetic wettability values. We developed the synthetic samples by initially assuming that all the pores were water-wet and contained only water. Then the hydrocarbon phase was assumed to migrate into the connected pores of the sample displacing all the water in the pore space. We identified the connected hydrocarbon-saturated pores by using a two-way travel 3D connected-component labeling algorithm (Purba et al. 2018). The rock was then assumed to age for a long time so that the wettability of all the connected pores changed from water- to hydrocarbon-wet and the hydrocarbon phase saturated the entire pore-space of these connected pores. The nonconnected pores in the sample were assumed to remain water-wet. **Fig. 2** shows the thin-section image slices of sample C3 demonstrating the effect of changes in rock wettability and fluid saturation in the pore space of one of the samples used in this study.

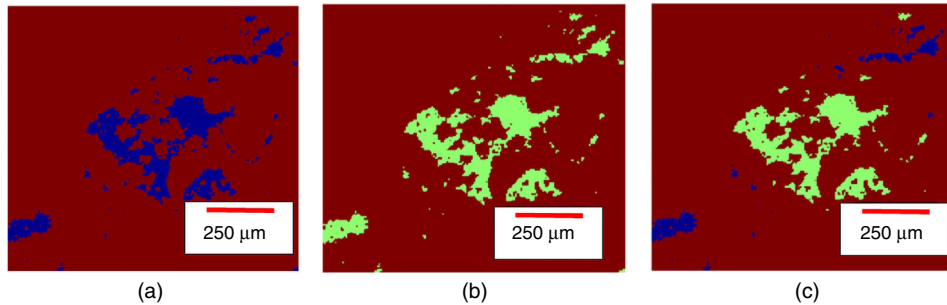


Fig. 2—Thin-section image slices of Sample C3 and synthetically modified sample CS3, demonstrating the fluid distribution in the sample when it is (a) water-wet, (b) hydrocarbon-wet, and (c) partially saturated mixed-wet. The red, blue, and green regions represent grains, water, and hydrocarbon phases, respectively.

Table 2 also lists the properties of the synthetic partially saturated mixed-wet samples created from the pore-scale samples. We estimated S_W and f_W directly from the pore-scale images of these synthetic samples. The value of f_{HC} was then calculated via

$$f_{HC} = 1 - f_W. \quad (11)$$

Next, we used Eq. 4 to estimate the pore-scale wettability values of all the partially saturated mixed-wet samples.

Pore-Scale Numerical Simulation and Inversion of Magnetization Decay. We modified an existing finite volume simulator OpenFOAM® (OpenCFD Ltd., Berkshire, UK) (Weller et al. 1998) to numerically solve Bloch-Torrey equations (Bloch 1946; Torrey 1956) with the mixed boundary condition given in Brownstein and Tarr (1979). More details on the numerical modeling approach can be found in the publications of Tandon et al. (2017) and Tandon and Heidari (2018). We used the Carr-Purcell-Meiboom-Gill (CPMG) sequence for simulating the magnetization decay in all the rock samples (Carr and Purcell 1954; Meiboom and Gill 1958). We assumed an initial constant magnetic field of 470 Gauss, which generates the initial magnetization at a Larmor frequency of 2 MHz. We assumed that the echo spacing time is small enough so the effects of internally generated and externally applied magnetic field gradients are negligible. **Table 3** summarizes the assumed properties of fluids and rocks used in the numerical simulations.

Parameter	Value	Parameter	Value
D_w (cm ² /s)	2.5×10^{-5}	ρ_{2WC} (μm/s)	8
D_{HC} (cm ² /s)	1.5×10^{-5}	ρ_{2HCC} (μm/s)	2
T_{BW} (seconds)	2.5	ρ_{2WS} (μm/s)	15
T_{BHC} (seconds)	1.5	ρ_{2HCS} (μm/s)	3

Table 3—Assumed input parameters to numerical simulations of Bloch-Torrey equations.

We first simulated the magnetization decay in all samples by assuming that they are water-saturated water-wet and hydrocarbon-saturated hydrocarbon-wet. These simulation results were used as model inputs for the new NMR-based wettability index. We then created the partially saturated mixed-wet samples and simulated magnetization decay in them. All the magnetization decay curves were inverted using one-dimensional (1D) Levenberg-Marquardt inversion method, with the zeroth order curve fitting scheme having 121 T_2 bins spaced between 0.1 milliseconds and 10 000 milliseconds (Aster et al. 2005). The 1D inversion kernel is given by

$$M(t) = \phi_T \sum \phi_i e^{-\frac{t}{T_{2i}}} \dots \dots \dots (12)$$

Calculating the NMR-Based Wettability Index in the Pore-Scale Domain. The final step is to calculate the NMR-based wettability index using the index expressed in Eq. 8. We used the inverted magnetization decay signals in all the samples to compute the NMR-based wettability index. We calculated T_{2HM} from the simulated T_2 distribution using Eq. 9 and T_{BHM} using Eq. 10 considering the water and hydrocarbon saturations given in Table 2 with fluid properties given in Table 3. We validated the new NMR-based wettability index by comparing it to pore-scale values of rock wettability calculated using Eq. 4.

Core-Scale Experimental Verification of the New NMR-Based Wettability Index. Fig. 3 summarizes the workflow used to experimentally verify the estimates of wettability using the new NMR-based wettability index. Table 4 shows the properties of the TC limestone outcrop core samples used for experimental verification of the NMR-based wettability index in the core-scale domain. The cores were cut to have the same dimensions and have porosity values between 0.26 and 0.29. We alter wettability of each of these core samples by injecting either brine, a solution of anionic surfactant in brine, or a solution of naphthenic acid in decane into these samples. The alteration in core wettability is quantified by measuring both the AH index and the contact angle of sessile drops. Then, we measure the magnetization decay of each sample at different water saturation states. We use the magnetization decay signals to obtain the T_2 distributions, which are then used to verify the applicability of the new NMR-based wettability index (Eq. 8) in the core-scale domain.

Altering the Wettability of the Core Samples. We used four TC limestone core samples, called TC-A, TC-B, TC-C, and TC-D, to verify the reliability of the introduced NMR-based wettability index. To alter the wettability of the core samples, we first dried them in an oven at a temperature of 80°C for 36 hours. Next, TC-A, TC-B, and TC-C samples were saturated with 2 wt% potassium chloride (KCl) brine. To saturate the core samples, we used a vacuum pump for 8 hours at −14 psi. Then, we introduced the brine into the vacuumed pore space. We kept the TC-A core sample in the brine solution while we altered the wettability of the remaining samples. To saturate and alter the wettability of the TC-D to hydrocarbon-wet, we used a vacuum pump to saturate it with a solution of 10 vol% of naphthenic acid in decane (Sweeney and Jennings 1960). Then, we aged TC-D sample at room temperature—in the naphthenic acid and decane solution—for 3 days so that its wettability changes to hydrocarbon-wet.

We altered the wettability of two core samples, TC-B and TC-C, to mixed-wet by using an anionic surfactant solution. We first injected a solution of 0.2 wt% anionic surfactant in 2 wt% KCl brine in the core samples using a core-flood setup. The anionic surfactant, sodium 1-decanesulfonate, has a negatively charged hydrophobic tail group that allows the surfactant to adsorb to the rock and alter the interfacial properties. We pumped at least ten pore volumes (PVs) in each sample at a flow rate of 1.0 cm³/min. The fluid injection was done with an axial confining pressure of 1,000 psi. Then, we aged TC-B in the anionic surfactant solution for three weeks so that it would become mixed-wet. In the case of TC-C sample, the wettability was further altered by injecting decane into the pore space. Finally, we aged the TC-C sample in decane for three weeks at room temperature to make it slightly hydrocarbon-wet (Li and Firoozabadi 2000).

Wettability Measurements in the Core Samples. We verified the alteration of rock wettability of the cores using two different methods. First, we qualitatively estimated the wettability by measuring the contact angle of sessile drops on the core sample surface (Young 1805; Anderson 1986b; Hirasaki 1991). In addition, we used the Amott method (Amott 1959; Sharma and Wunderlich 1985) to quantitatively measure the wettability of the core samples.

In the case of the sessile drop method, we placed the core sample in a rectangular jar and filled it with decane. Then, we used a pipet to place a drop of brine on the core sample surface. A Ramé-Hart goniometer setup was used to take a high-resolution picture of the sessile drop. We repeated the sessile drop experiment several times for each core sample. We measured the angle between the core sample and the drop of brine using the On-Screen Protractor Java application (Straffi 2012). A water-wet rock has a smaller contact angle between the brine and the rock surface as the brine clings to the grains. Contrarily, a hydrocarbon-wet rock will have a larger contact angle as the brine is repelled from the grains. Mixed-wet rocks can have a wider range of contact angles, which depend on distribution of water- and hydrocarbon-wet grains (Neumann and Good 1979).

In addition, we measured the spontaneous imbibition of brine and the spontaneous drainage of decane to quantify rock wettability using the AH index (Amott 1959). Each core sample was saturated with decane using the core-flood setup to reach the irreducible water saturation, S_{WIRR} . Then, we placed the core sample in an upright Amott cell. We covered the sample with brine and measured the volume of decane that is spontaneously replaced by brine to calculate S_{SPW} . Next, we used a centrifuge to forcefully imbibe the brine into the samples until they reached the residual oil saturation, S_{OR} . The centrifuge was run at a speed equivalent to a capillary pressure of 15 psi. For the TC-D sample, we investigated the residual oil saturation using a capillary pressure of 150 psi as the sample was hydrocarbon-wet. Then, the inverted Amott cells were used to measure the spontaneous imbibition of decane, S_{WSPD} . Finally, we forcefully injected decane using the centrifuge at a capillary pressure of 15 psi until we reached the irreducible water saturation, S_{WIRR} . Eq. 1 was used to estimate the AH wettability index of the core samples.

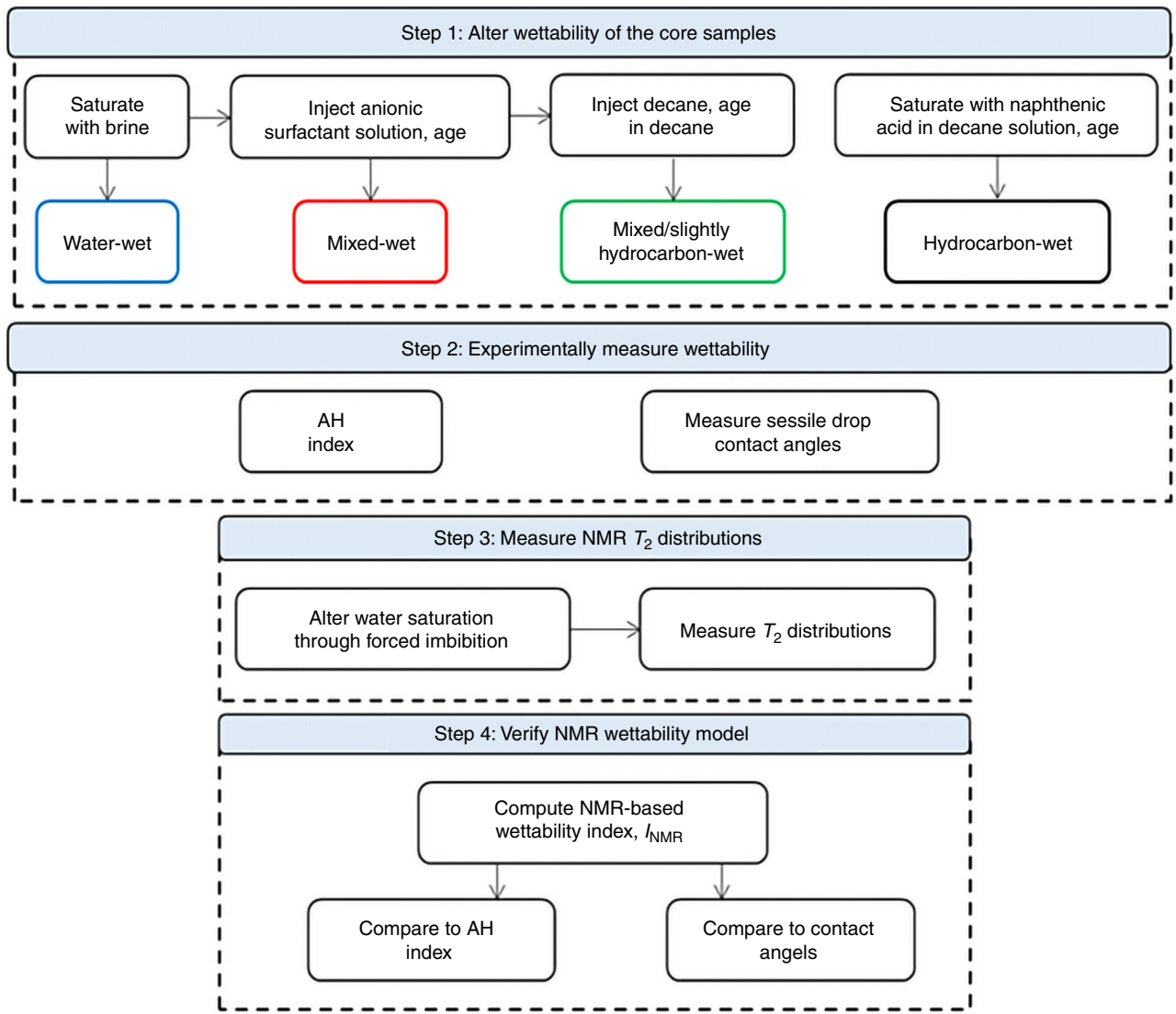


Fig. 3—Workflow for core-scale experimental verification of the introduced NMR wettability index.

Sample	Diameter (cm)	Length (cm)	ϕ_T	Injected Fluids
TC-A	3.76	5.20	0.26	Brine
TC-B	3.76	5.18	0.24	Brine, surfactant solution, aging process
TC-C	3.76	5.18	0.26	Brine, surfactant solution, decane, aging process
TC-D	3.77	5.14	0.29	Naphthenic and decane solution, aging process

Table 4—Summary of the properties of the TC-A, TC-B, TC-C, and TC-D core samples and the injected fluids for wettability alteration.

Obtaining Fully Water-Saturated Water-Wet and Hydrocarbon-Saturated Hydrocarbon-Wet Model Input Parameters. We performed the T_2 measurements on the hydrocarbon-wet TC-D sample when it was fully saturated with decane to obtain T_{2HCHM} . Likewise, we measured the NMR T_2 response on the water-wet TC-A sample when it was fully water-saturated to obtain T_{2WHM} .

Preparing Partially Water-Saturated Core Samples. We reached different saturation states after spontaneous (Amott cell) and forced (centrifuge) imbibition of brine and decane in the core samples. At each stage, we stopped the centrifuge and measured the mass of the partially water-saturated core sample. As we changed the water saturation of the sample, we weighed the sample and gravimetrically estimated the water saturation via

$$S_W = \frac{M_S - M_D - (V_P \times \text{den}_{HC})}{V_P(\text{den}_W - \text{den}_{HC})} \quad (13)$$

where M_D is the dry mass, M_S is the saturated mass, V_P is the PV, den_{HC} is the density of hydrocarbon, and den_W is the density of brine.

NMR Measurements on Partially Saturated Cores and Inversion of Magnetization Decay. We used a benchtop 2 MHz-frequency NMR spectrometer to measure NMR responses of the rock samples in both partially saturated and fully saturated core samples. All the measurements were performed at room temperature and atmospheric pressure. The CPMG pulse sequence was used to obtain the T_2 response in each core sample. We used an echo time of 100 microseconds, 80,000 echoes, and a minimum signal-to-noise ratio of 200 for acquiring the magnetization decay in the CPMG sequence. We background-corrected all the NMR T_2 responses by first running the measurements in the dried core samples covered with saran wrap. Then, we subtracted the magnetization decay obtained from these measurements with those obtained for fluid-saturated core samples. This limits the influence of any background noise and allows us to accurately quantify the influence of fluid saturation and rock wettability from NMR measurements. The magnetization decay was inverted using Eq. 12 with 1D Levenberg-Marquardt inversion method and using the zeroth order curve fitting scheme with 121 T_2 bins spaced between 0.1 milliseconds and 10 000 milliseconds.

Calculating NMR-Based Wettability Index in the Core-Scale Domain. The final step is to calculate the NMR-based wettability index as expressed in Eq. 8. We calculated T_{2HM} from the measured T_2 distribution using Eq. 9 and calculated T_{BHM} using Eq. 10 considering the actual water saturation in each core sample estimated using gravimetric measurements. The constant values of T_{BW} and T_{BHC} were measured to be 2610 milliseconds and 1443 milliseconds, respectively. We normalized the magnetization decay of each sample by its total porosity to limit the influence of porosity variation between different samples.

Results

Pore-Scale Numerical Validation of the New NMR-Based Wettability Index. Fig. 4 shows the simulated T_2 distributions in one of the rock samples demonstrating the influence of both wettability and fluid saturation on NMR T_2 responses. Similar results were obtained for all the other rock samples and are presented in Appendix B. The locations of T_2 peaks are correlated to both wettability and fluid saturation of the rock sample. In accordance with Eq. 8, changes in fluid saturation and rock wettability cause a shift in the locations of the T_2 peaks in all the rock samples. In all the samples, the simulated T_2 distributions of partially saturated mixed-wet samples lie in the middle of the simulated T_2 distributions for the fully water-saturated water-wet and fully hydrocarbon-saturated hydrocarbon-wet samples.

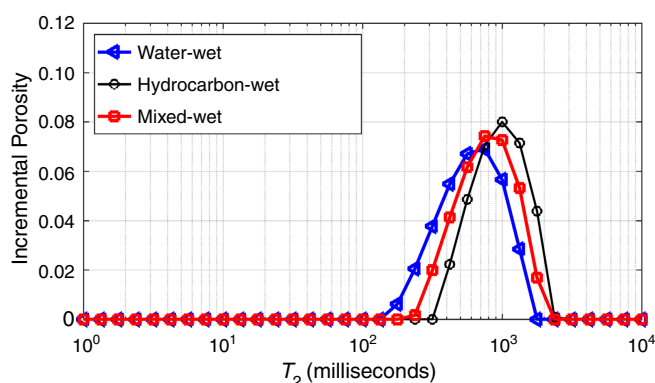


Fig. 4—Sample C4: Simulated T_2 distribution, showing the influence of wettability and fluid saturation on NMR responses. The blue, black, and red curves show the T_2 distribution for fully water-saturated water-wet, fully hydrocarbon-saturated hydrocarbon-wet, and partially saturated mixed-wet rock samples, respectively.

Fig. 5 quantifies the variation in the arithmetic mean of T_2 peaks (T_{2AM}) for all the samples as rock wettability changes from hydrocarbon- to water-wet. Fig. 5 shows that the arithmetic mean of T_2 peaks decreases as the rock samples become more water-wet demonstrating an increase in the T_2 relaxation rate. A similar increase in the T_2 relaxation rate with increasing water wettability has been reported in previous publications by Brown and Fatt (1956), Hsu et al. (1992), and Guan et al. (2002).

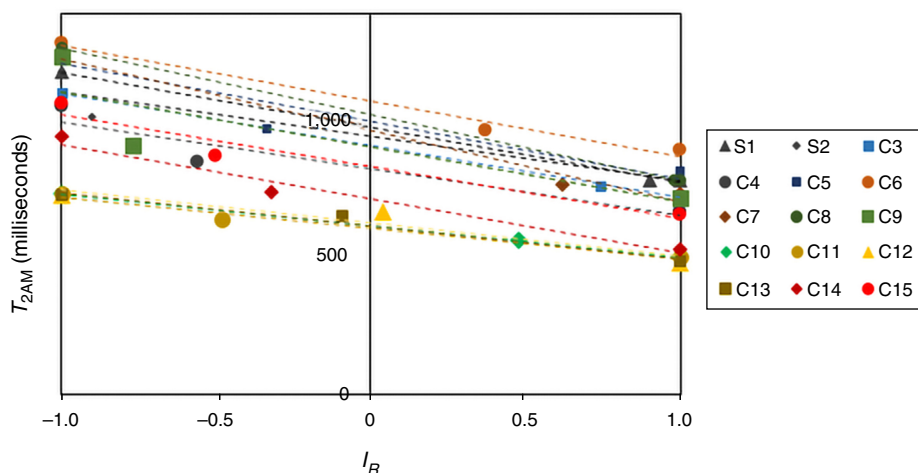


Fig. 5—Variation in the arithmetic mean of T_2 peaks (T_{2AM}) as the pore-scale rock wettability (I_R) of the synthetic partially saturated mixed-wet samples changes from -1 to 1 .

Fig. 6 and Table 5 show the comparison of the pore-scale rock wettability of the synthetic partially saturated mixed-wet samples calculated using Eq. 4 against the NMR-based estimates of wettability index calculated using Eq. 8. We observe a good agreement between the two wettability indices with the average relative difference of 20% in estimates of rock wettability.

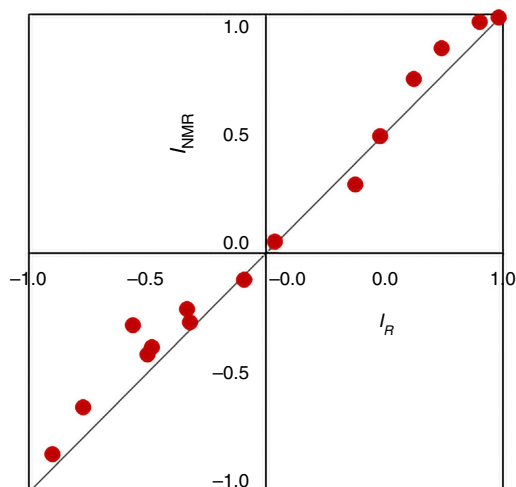


Fig. 6—Comparison of the NMR-based estimates of wettability index (I_{NMR}) from simulated T_2 response against the pore-scale rock wettability (I_R) of the synthetic partially saturated mixed-wet samples.

Sample Name	Pore-Scale Rock Wettability (I_R)	NMR-Based Wettability (I_{NMR})	Sample Name	Pore-Scale Rock Wettability (I_R)	NMR-Based Wettability (I_{NMR})
S1	0.90	0.97	C9	-0.78	-0.64
S2	-0.90	-0.84	C10	0.48	0.49
C3	0.74	0.86	C11	0.32	0.44
C4	-0.56	-0.30	C12	0.04	0.05
C5	-0.32	-0.23	C13	-0.08	-0.12
C6	0.38	0.29	C14	-0.32	-0.29
C7	0.62	0.73	C15	-0.50	-0.42
C8	0.98	0.99			

Table 5—Summary of the pore-scale rock wettability (I_R) and NMR-based estimates of wettability (I_{NMR}) on the synthetic partially saturated mixed-wet samples S1 to C15.

Core-Scale Experimental Validation of the New NMR-Based Wettability Index. In this section, we present the wettability alteration results and the NMR-based wettability indices of the TC core samples. The wettability alteration of the core samples from water-wet to mixed-wet and hydrocarbon-wet was confirmed using both the AH index and sessile drop contact angle measurements. The water saturation at each stage in each core sample was estimated using gravimetric measurements. The T_2 distributions were measured for each core sample for water saturation states ranging from 0.27 to 0.61. Finally, we compared the calculated I_{NMR} values to the measured AH index and sessile drop contact angles.

Validating the Core-Scale Wettability Alteration Using AH and Contact Angle Measurements. To assess the contact angle measurements of the cores, we used the goniometer setup. We measured the contact angle between the brine and rock three times and verified it within an accuracy of 3° . **Table 6** summarizes the measured contact angles which range from 21° for the water-wet rock to 153° for the hydrocarbon-wet rock. **Table 7** summarizes water saturations used to calculate I_{AH} for the all core samples using Eq. 1. We estimated water saturation errors by repeating the gravimetric measurements and observed them to be within a range of 3% for the samples used in this study. For these core samples, the sessile drop contact angle measurements and AH index are in agreement when describing the wettability of each sample.

Sample	Contact Angles (degrees)	Summary
TC-A	21–36°	Water-wet
TC-B	48–101°	Mixed-wet
TC-C	72–97°	Mixed/slightly hydrocarbon-wet
TC-D	90–153°	Hydrocarbon-wet

Table 6—Wettability assessment using sessile drop contact angle measurements.

Sample	S_{WIRR}	S_{WSP}	$1-S_{OR}$	S_{WSPO}	I_{AH}
TC-A	0.38	0.47	0.53	0.51	0.5
TC-B	0.29	0.38	0.60	0.57	0.2
TC-C	0.50	0.56	0.69	0.65	0.1
TC-D	0.27	0.30	0.66	0.43	-0.6

Table 7—Measured water saturations used to calculate the AH index

NMR Measurements of the Core Samples. Figs. 7a and 7b show the T_2 distributions with a water T_2 peak at 38 milliseconds and hydrocarbon T_2 peak at 562 milliseconds, respectively.

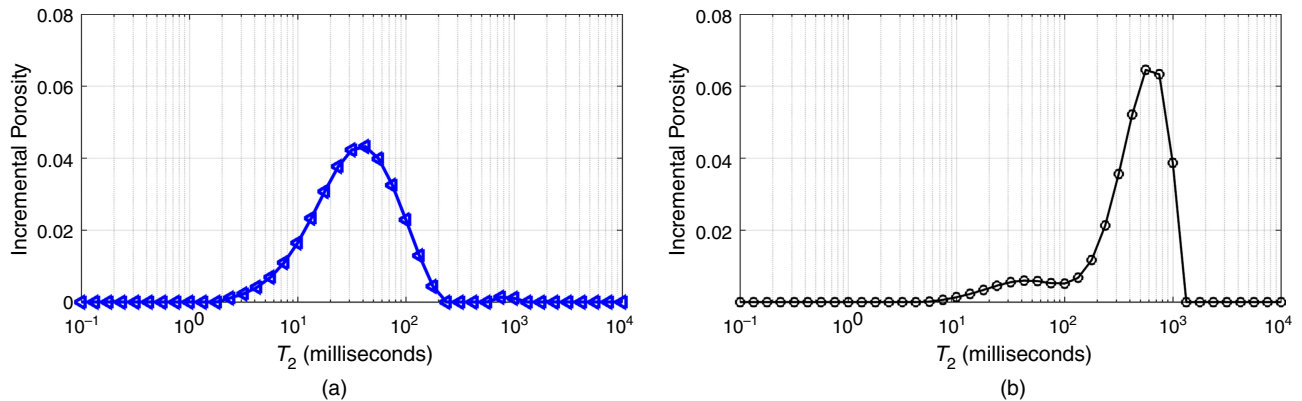


Fig. 7—NMR T_2 distribution of the calibration curves for (a) water-saturated water-wet sample, TC-A and (b) hydrocarbon-saturated hydrocarbon-wet sample, TC-D.

Fig. 8 displays the NMR T_2 distribution curves of each core sample at varying water saturations. Appendix C lists the water saturation values and the corresponding water and hydrocarbon T_2 distribution peaks for each core sample. We observe multimodal T_2 distributions due to the presence of both water and hydrocarbon phases in the core samples. By comparing Fig. 8 against the water and hydrocarbon curves in Fig. 7, we can attribute the fast-relaxing peaks as the water phase and the slow relaxing peaks as the hydrocarbon phase in the core samples. The values of water-dominant peaks vary between 31 and 38 milliseconds and the values of hydrocarbon-dominant peaks vary between 908 and 1079 milliseconds.

Fig. 9 displays the NMR T_2 distribution curves of core samples TC-A, TC-B, TC-C, and TC-D at similar water saturation states (between 0.52 and 0.54). All the core samples show a multimodal T_2 distribution due to the copresence of water and hydrocarbon phases. The dominant water peak for the hydrocarbon-wet TC-D sample is not as clear as the other samples but could be identified from the inflection point in the T_2 distribution curve. Fig. 9 shows that the dominant hydrocarbon peak shifts left toward faster relaxing times, as the rock becomes more hydrocarbon-wet.

Core-Scale Verification of the Reliability of the New NMR-Based Wettability Index. We used the harmonic mean of the T_2 distributions, T_{2HM} and T_{BHM} to calculate the NMR-based wettability index for each core sample at different saturation states. Appendix C also presents the T_{2HM} and T_{BHM} calculated for each core sample at different water saturations shown in Fig. 8. Fig. 10a compares the calculated NMR-based wettability with the experimentally measured AH index. The average relative difference in estimates of rock wettability using the aforementioned methods is 44%. Fig. 10b compares the calculated NMR-based wettability index against the average contact angle of each core sample. The measured contact angles provide qualitative estimates of the overall wettability of the rock samples and helps to verify the applicability of the new NMR-based wettability index in mixed-wet rock systems. Table 8 summarizes the measurements of sessile drop contact angle, the AH index, and the average NMR-based wettability index of each core sample.

Comparison of the Performance of the New NMR-Based Wettability Index Against the Previously Developed NMR-Based Wettability Indices. Fig. 11 compares the results obtained from applying the new index given by Eq. 8, Fleury and Deflandre (2003) index (I_{FD}) given in Eq. 5, and Chen-Hirasaki-Flaum index (I_{CHF}) given in Eq. 7 to the NMR T_2 responses shown in Fig. 8. An average relative difference of 44% was observed between the wettability estimates from the new NMR-based wettability index and those obtained from the AH method. The average relative difference of 173 and 122% was observed between the wettability estimates from the AH method and those obtained from Fleury and Deflandre (2003) and Chen et al. (2006), respectively. These errors in wettability estimates are higher for the previously introduced NMR-based indices (I_{FD} and I_{CHF}) because both these models were derived for rocks with unimodal pore structure. Moreover, both these models were reliably verified only in Berea Sandstones, which has a unimodal pore-size distribution. The pore-throat size distribution obtained for the TC samples is multimodal (Garcia et al. 2018). This is also indicated by the NMR T_2 distributions shown in Fig. 7, which shows a clear distinction between the pores that might be saturated by water and hydrocarbon phases. Hence, applying the previously derived NMR-based wettability indices (I_{FD} and I_{CHF}) yields higher errors than those obtained by applying of the new NMR-based wettability index.

Furthermore, both Fleury and Deflandre (2003) and Chen et al. (2006) discuss that the NMR-based wettability indices developed in those publications should be applied only when T_2 measurements are performed at irreducible water and residual oil saturations. Applying these previously developed wettability indices at fluid saturation states other than irreducible water and residual oil saturations might not provide accurate estimates of rock wettability for partially saturated mixed-wet rocks.

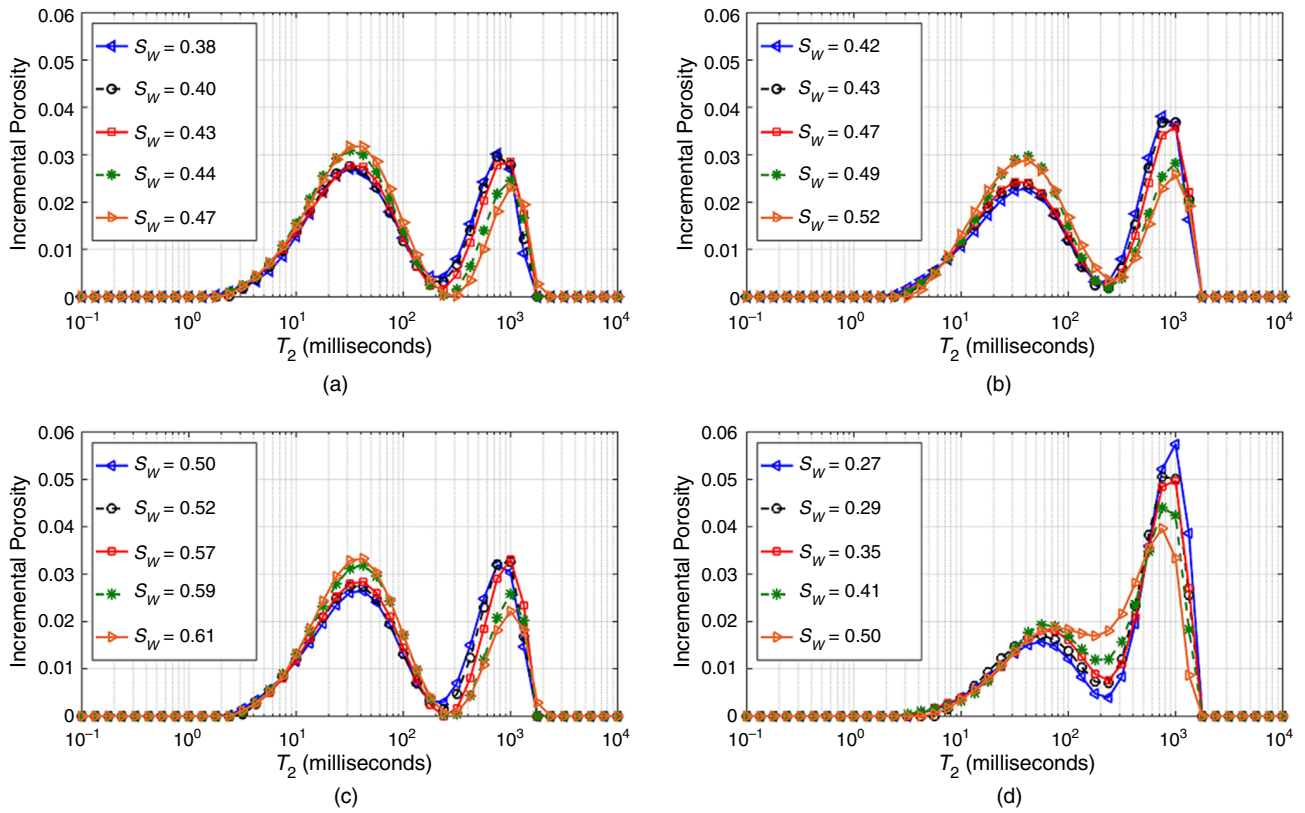


Fig. 8—NMR T_2 distributions at varying water saturations for rock samples (a) TC-A, (b) TC-B, (c) TC-C, and (d) TC-D.

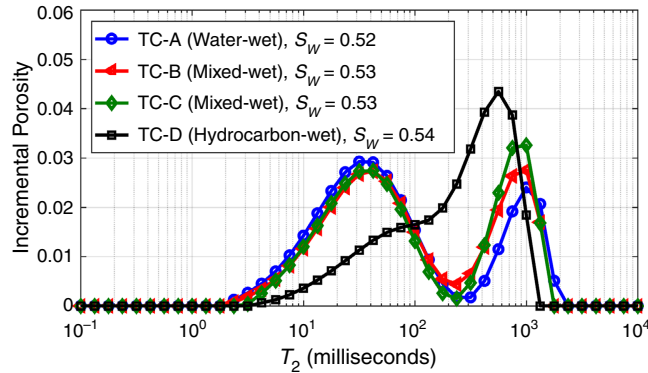


Fig. 9—NMR T_2 distributions at varying rock wettability for the selected rock samples. The blue, red, green, and black curves show T_2 distributions for rock samples TC-A, TC-B, TC-C, and TC-D, respectively.

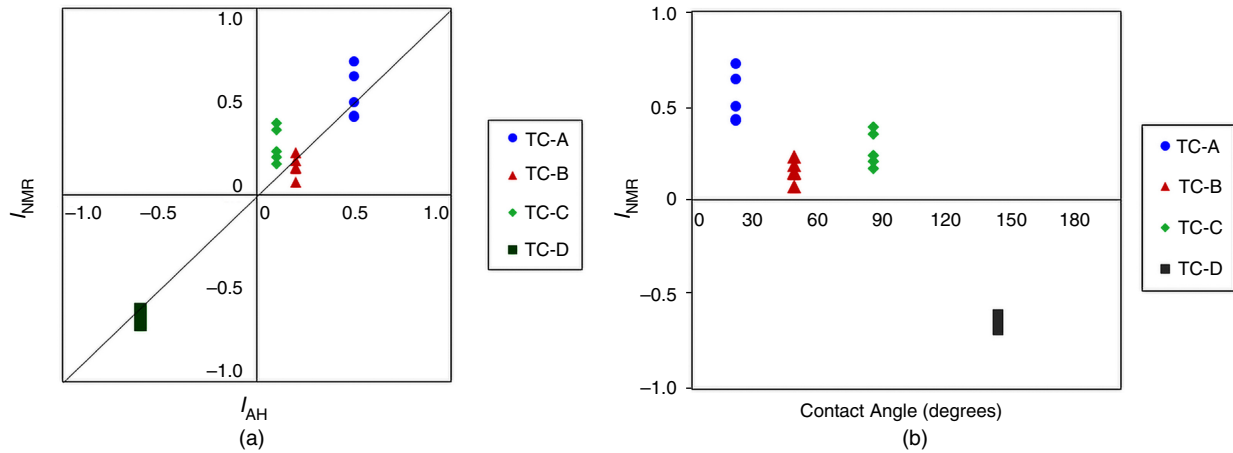


Fig. 10—Comparison of experimentally measured (a) AH index (I_{AH}) and (b) contact angle against the new NMR-based wettability index (I_{NMR}) measured for five different water saturation levels. The blue circles, red triangles, green diamonds, and black squares represent core samples TC-A, TC-B, TC-C, and TC-D, respectively.

Sample	Contact Angle (degrees)	I_{AH}	I_{NMR}
TC-A	21–36°	0.5	0.51
TC-B	48–101°	0.2	0.16
TC-C	72–97°	0.1	0.27
TC-D	90–153°	−0.6	−0.66

Table 8—Summary of the mean rock wettability indices for core samples TC-A, TC-B, TC-C, and TC-D.

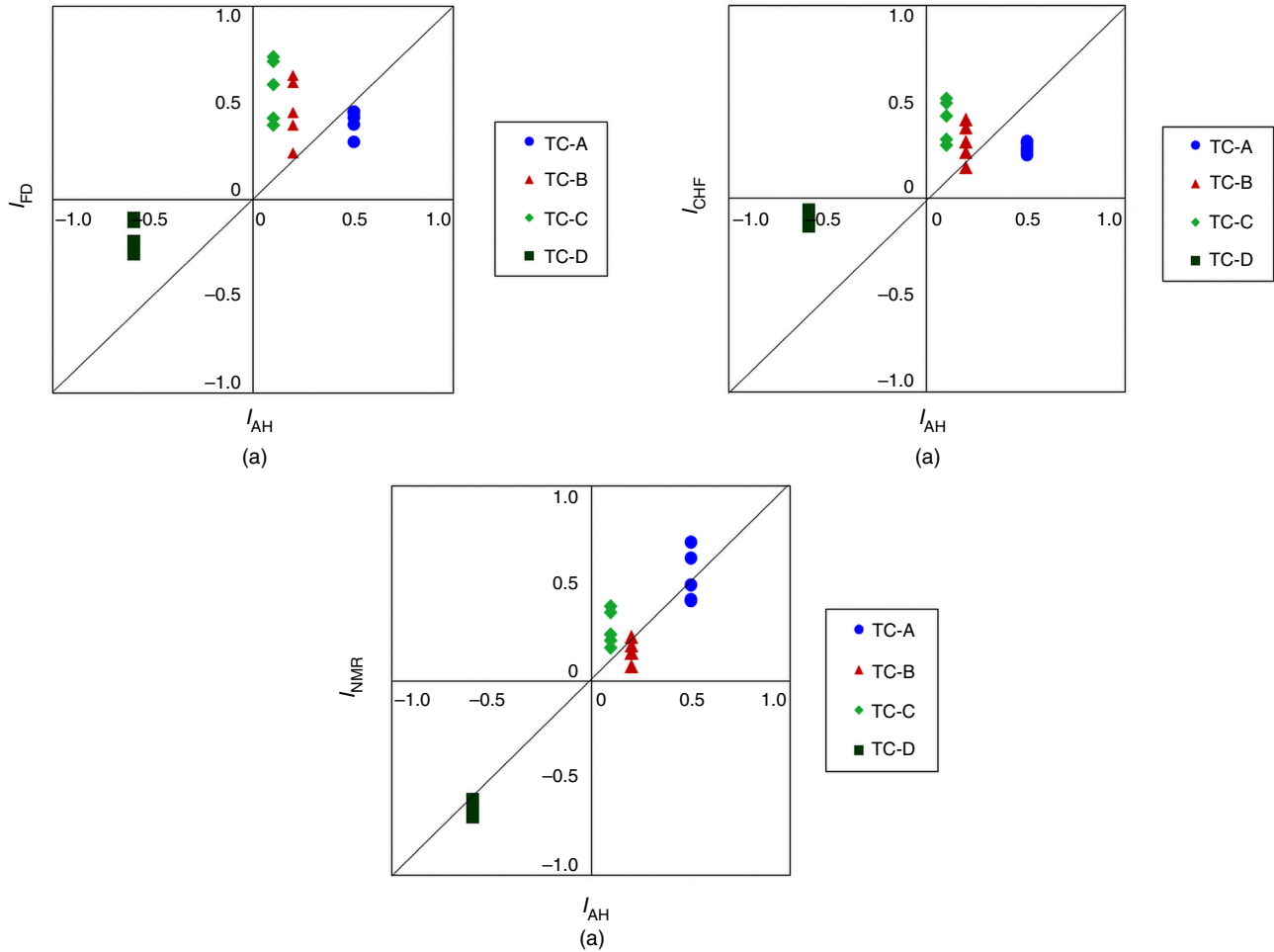


Fig. 11—Comparison of the AH index against the NMR-based wettability models introduced in (a) Fleury and Deflandre (2003), (b) Chen et al. (2006), and (c) this manuscript (I_{NMR}).

Discussion

The derivation of the new NMR-based wettability index in Eq. 8 indirectly incorporates the effect of complex pore-structure and fluid saturation in quantifying rock wettability by including the T_2 distributions of fully water-wet and fully hydrocarbon-wet rock samples. Therefore, the new index does not require the pore geometry to be explicitly specified in estimating the wettability of the rock sample.

The derivation of the NMR-based wettability index assumes limited influence of internal magnetic field gradients on the T_2 measurements. The influence of magnetic field gradients is proportional to the square of echo spacing time during the CPMG sequence. We have performed all the T_2 measurements at the lowest possible echo spacing time available in the spectrometer. In addition, TC has been shown in previous publications to contain mostly calcium carbonate (more than 85%) and insignificant concentrations of paramagnetic clays (Mowar et al. 1994). Calcium carbonate has magnetic susceptibility close to that of formation brine and does not impose significant internal magnetic field gradients (AlGhamdi and Arns 2012). Hence, the influence of internal magnetic field gradients is considered to be negligible on our NMR T_2 measurements.

We also assume that all the pores in the sample are in the fast diffusion regime (Brownstein and Tarr 1979) with limited diffusive coupling among them. The assumption of fast diffusion regime with limited influence of diffusive coupling among pores is made for all NMR-based wettability models including those published in Fleury and Deflandre (2003), Chen et al. (2006), Looyestijn and Hofman (2006), and Al-Muthana et al. (2012). This assumption has been shown to remain valid for most sandstone and carbonate rock samples (Kleinberg and Horsfield 1990; Kleinberg et al. 1994; Foley et al. 1996), including those used in this paper.

Similar to previous publications, the pore surfaces are assumed to have uniform values of water surface relaxivity. The values of water surface relaxivity depends on the surface concentration of paramagnetic particles and the residence time of protons on the pore surface (Foley et al. 1996; Korb 2011) and can be assumed to be uniform for grain surfaces. The values of hydrocarbon surface relaxivity might depend on the strength of dipolar coupling, proton interaction with paramagnetic rock or asphaltene components, and the hydrocarbon molecule residence times. Hence, the hydrocarbon surface relaxivity could vary depending on its molecular mass and asphaltene composition (Godefroy et al. 2002; Looyestijn 2008). However, hydrocarbons with low asphaltene concentration and small molecular mass display constant values of surface relaxivity (Chen et al. 2006; Rios et al. 2014). The effects of water and hydrocarbon surface relaxivities are also accounted for by using the NMR T_2 responses of fully water-wet and hydrocarbon-wet samples of the same rock type.

Results demonstrated in Fig. 6 and Table 6 show that the new NMR-based wettability index correlates well with pore-scale rock wettability with coefficient of determination of 0.98 and the average relative difference of 20%. Fig. 10 and Table 9 show an average relative difference and coefficient of determination observed between NMR-based wettability index and the AH index were 44% and 0.92, respectively. We observed a better correlation between the estimated NMR-based wettability index and the pore-scale rock wettability compared to that obtained in the core-scale domain. There could be two reasons for this observation. The main reason is that both the pore-scale and the NMR-based wettability indices quantify the fractional surface areas covered by different fluids in the rock sample. As both the indices quantify the same underlying rock property, we observe an excellent correlation between these indices. On the other hand, the AH index is determined via a displacement test that is based on quantifying endpoint water and hydrocarbon saturation values. These endpoint saturation values could be affected by rock properties other than wettability such as rock permeability and saturation history of the sample. Hence, the correlation between the AH index and the NMR-based wettability index is not as good as that of the pore-scale domain.

The other reason could be the process of creating synthetic mixed-wet partially saturated pore-scale samples, which considers an ideal situation in which the hydrocarbon phase migrates into the connected pores of the sample. Then, we assumed that all the surfaces of all pores saturated with the hydrocarbon phase become completely hydrocarbon-wet with a uniform value of hydrocarbon surface relaxivity. In the core-scale domain, the fluid distribution in the sample depends not only on rock wettability, but is also affected by pore-size distribution, saturation history, and distribution of connected fluids in the core that has not been taken into account in the process of creating synthetic mixed-wet pore-scale samples.

The advantage of this new NMR-based wettability index is that it can be used to quantify rock wettability at any saturation state, while the previously derived NMR-based indices such as I_{FD} and I_{CHF} can be used to quantify rock wettability only at the residual hydrocarbon and irreducible water saturations. Hence, the new NMR-based wettability index can be easily applied to log-scale measurements for any conducted in-situ conditions. In addition, previously introduced NMR-based wettability indices (I_{FD} and I_{CHF}) rely on selecting a single T_2 peak to account for the size of dominant pores in rock samples. This limits their application to rocks with unimodal pore structures because the existence of small peaks due to micropores in rocks are disregarded in estimates of rock wettability. The new NMR-based wettability index can be applied to rocks with multimodal pore-size distributions because the harmonic mean accounts for both small and large pores.

The advantage of using the new index over the inversion algorithms proposed in the publications of Looyestijn (2008) and Al-Muthana et al. (2012) is that we accurately account for pore surface area distribution. In addition, we limit the errors that might be introduced by assuming different functions for determining the saturations and wettability states of individual pores as proposed in the algorithm by Looyestijn and Hofman (2006). Finally, applying the new NMR-based wettability index is faster than the previously published models as we do not need to carry out an optimization process to obtain the wettability of mixed-wet partially saturated rocks.

The limitations of the new NMR-based wettability index are that we need to obtain water and hydrocarbon saturations before we can quantify the rock wettability. In this study, we estimated fluid saturation using image analysis in the pore-scale domain and gravimetric measurements in the core-scale domain. Uncertainty in gravimetric measurements could be introduced due to excess moisture on the sample, or due to evaporation of the fluids. We mitigated this uncertainty in measurements by storing all the samples in saran wrap when they are not being used. We blotted the exterior of the cores with a premoistened paper towel to ensure that the gravimetric measurements are consistent throughout experiments.

In addition, like all the previously introduced NMR-based wettability indices, we are comparing the NMR responses for rocks at different saturation states in T_2 space. Hence, the assumptions of the inversion algorithm such as the range and number of T_2 bins might affect the estimation of rock wettability. For applying the new NMR-based wettability index, we recommend using normalized NMR responses, NMR data with high signal-to-noise ratio, constant range and number of T_2 bins, and an optimized value of the regularization parameter.

Sensitivity of the New NMR-Based Wettability Index to Water Saturation. Fig. 12 shows the variation of the new NMR-based wettability index for the range of water saturations used in the experiments. The water-wet TC-A and the hydrocarbon-wet TC-D core samples have the most extreme values of estimated new wettability index. In the case of TC-A, we estimate the new wettability index ranging from 0.38 to 0.67 for water saturations of 0.38 to 0.47. However, TC-D has a narrower range of wettability indices from -0.61 to -0.68 for water saturations of 0.25 to 0.50.

Although water saturation is taken into account in the new model, we still observe a measurable sensitivity of the estimates of wettability from the new NMR-based wettability index to water saturation, particularly for the water-wet sample. Possible reasons for this sensitivity are discussed as follows. The first reason can be the assumption of fast diffusion of proton spins in the pore space. This assumption for fast diffusion might be violated in some large vugs that might be present in the core samples. In addition, the new NMR-based wettability model does not take into account the heterogeneity of surface relaxivities, and diffusive coupling among pores. Finally, the uncertainty in laboratory measurements for water saturation assessment can contribute to this sensitivity. Fig. 12 shows that even though there is some uncertainty in saturation assessment and some of these assumptions might not strictly hold true, the model performs very well in quantifying the wettability of the carbonate samples that are used in this study.

Proposed Workflow for Field Application of the New NMR-Based Wettability Index. Use of the new NMR-based wettability index for wettability assessment requires knowledge of fluid saturations and their bulk relaxivity values. In application to the log-scale domain, measurements such as resistivity, which is sensitive to both fluid saturation and rock wettability, can be used to estimate water saturation. However, due to the impact of wettability, resistivity models such as Archie's (1942) might require extensive calibration to be directly applied to mixed- or oil-wet (OW) rocks. In such cases, Valori et al. (2017) described a simple method on the basis of $D-T_2$ that could be used to obtain water saturation. In addition, resistivity models such as those discussed in Newgord et al. (2018) that incorporate wettability could be used to reliably estimate fluid saturation in mixed-wet rocks.

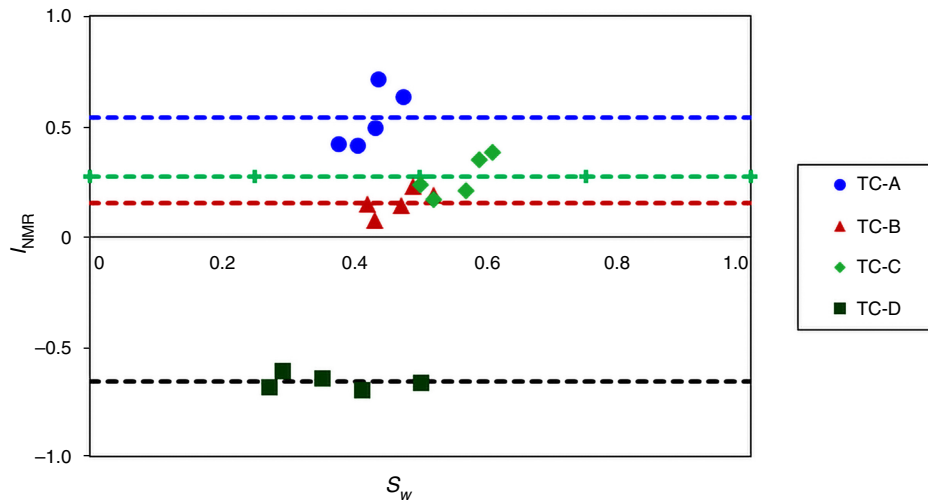


Fig. 12—Sensitivity analysis on the influence of water saturation on the estimates of the new NMR-based wettability index. The blue, red, green, and black symbols represent core samples TC-A, TC-B, TC-C, and TC-D, respectively. The symbols show the NMR-based wettability index for different water saturation levels, and the dashed lines demonstrate the average values of the new NMR-based wettability index for each sample.

The bulk relaxivity values for water and hydrocarbon phases can be evaluated using the experiments or theoretical models, such as those published in Hirasaki et al. (2003). The fully water-saturated water-wet and fully hydrocarbon-saturated hydrocarbon-wet T_2 responses can be obtained in the laboratory by modifying the wettability of extracted core samples of each rock type in the formation. Also, these responses could be obtained by using numerical simulations of brine and hydrocarbon NMR responses in the micro-CT images of rock samples such as those performed in the pore-scale section of this study.

NMR T_2 measurements are adversely affected by the internal magnetic gradients because of tool features or because of the presence of paramagnetic clay minerals (Morriss et al. 1997; Zhang et al. 2001, 2003; Tandon and Heidari 2018). If the NMR T_2 response is suspected to be influenced by internal magnetic field gradients, then the reliability of rock wettability assessment can be improved by using tools with smaller echo spacing times, using longitudinal response, and/or numerical corrections for the impacts of magnetic field gradients (Zhang et al. 2001, 2003; Sun and Dunn 2002; Tandon and Heidari 2018).

Conclusions

We introduced and verified the applicability of a new NMR-based wettability index that can be used for estimating wettability in partially water-saturated mixed-wet core samples. This new model enables wettability assessment in partially saturated rocks with complex pore-structure and multimodal pore-size distribution. The contributions of the new NMR-based wettability index over previously published NMR-based methods are listed as follows:

- The new NMR-based wettability index is applicable to core samples at different wettability and fluid saturation states.
- Unlike previously developed models, the new NMR-based wettability index does not require NMR measurements at irreducible water and residual hydrocarbon saturations.
- Unlike previously proposed NMR-based inversion algorithms, the new NMR-based wettability index accurately accounts for pore-surface area distribution and is significantly less computationally expensive.
- The new NMR-based wettability index is significantly faster than core-flood and centrifuge measurements (i.e., required for USBM and AH index wettability quantification methods), which might take weeks to measure for each rock sample.

The new NMR-based wettability index shows coefficients of determination of 0.98 and 0.92 when compared with rock wettability in the pore-scale domain and AH indices in the core-scale domain, respectively, confirming its applicability in both pore- and core-scale domains. Results demonstrated that the new index improves NMR-based wettability estimation when compared to previously developed NMR-based wettability indices as it accurately accounts for pore-surface area distribution, fluid saturation, and multimodal pore structure of rocks, which has been ignored in previously documented models.

Nomenclature

- A_D = area under the capillary pressure curves at the end of the drainage cycle
 A_{HCi} = surface area of hydrocarbon in pore i , m^2
 A_I = area under the capillary pressure curves at the end of the imbibition cycle
 A_P = total pore area, m^2
 A_{Wi} = surface area of water in pore i , m^2
 C_{WH} = ratio of water and hydrocarbon transverse surface relaxivity
 D = diffusivity, m^2/s
 D_{HC} = diffusivity of hydrocarbon phase, m^2/s
 D_W = diffusivity of water phase, m^2/s
 den_{HC} = density of hydrocarbon, kg/m^3
 den_W = density of brine, kg/m^3
 f_{HC} = fractional surface area of hydrocarbon phase in the rock sample
 f_W = fractional surface area of water phase in the rock sample
 I_{AGM} = Al-Mahrooqi-Grattoni-Muggeridge NMR-based wettability index
 I_{AH} = Amott-Harvey wettability index

I_{CHF} = Chen-Hirasaki-Flaum NMR-based wettability index
 I_{FD} = Fleury and Deflandre NMR-based wettability index
 I_{NMR} = new NMR-based wettability index
 I_R = pore-scale rock wettability index
 I_{USBM} = United States Bureau of Mines wettability index
 $M(t)$ = magnetization signal, p.u.
 M_D = dry mass of the core, kg
 M_S = saturated mass of the core, kg
 R = radius of the pore, m
 $S_i(r)$ = saturation of water phase in pore i
 S_{OR} = residual oil saturation
 S_W = water saturation
 S_{WIRR} = irreducible water saturation
 S_{WSP} = water saturation at end of spontaneous imbibition cycle
 S_{WSP0} = water saturation after spontaneous drainage cycle
 T = time, seconds
 T_{BHC} = bulk relaxation constant of hydrocarbon, milliseconds
 T_{BHM} = harmonic mean of bulk relaxation, milliseconds
 T_{BW} = bulk relaxation constant of water, milliseconds
 T_2 = transverse relaxation, milliseconds
 T_{2AM} = arithmetic mean of T_2 distribution of partially saturated mixed-wet sample, milliseconds
 T_{2HCHM} = harmonic mean of T_2 distribution of hydrocarbon-saturated hydrocarbon-wet sample, milliseconds
 T_{2HCi} = T_2 value of hydrocarbon phase in pore i , milliseconds
 T_{2HCP} = dominant peak of hydrocarbon phase T_2 distribution, milliseconds
 T_{2HM} = harmonic mean of T_2 distribution of partially saturated mixed-wet sample, milliseconds
 T_{2i} = decay time of pore i , milliseconds
 T_{2ij} = decay time of fluid component j in pore i , milliseconds
 T_{2ij} = decay time of fluid j located in pore i , milliseconds
 $T_{2LM^{or}}$ = log mean of the T_2 distribution at residual oil saturation, milliseconds
 $T_{2LM^{wc}}$ = log mean of the T_2 distribution at irreducible water saturation, milliseconds
 T_{2SHC} = harmonic mean of hydrocarbon surface relaxation, milliseconds
 T_{2SW} = harmonic mean of water surface relaxation, milliseconds
 T_{2WHM} = harmonic mean of T_2 distribution of the water-saturated water-wet sample, milliseconds
 T_{2Wi} = T_2 value of water phase in pore i , milliseconds
 T_{2WP} = dominant peak of water phase in the T_2 distribution, milliseconds
 V_{HCi} = volume of hydrocarbon in pore i , m³
 V_P = total pore volume, m³
 V_{Wi} = volume of water in pore i , m³
 $W_i(r)$ = fraction surface area of pore i in contact with water phase
 α_i = normalized pore area fraction of pore i
 α_{ij} = normalized pore area fraction of fluid j located in pore i
 ρ_{2HCC} = transverse surface relaxivity of hydrocarbon phase in carbonate, m/s
 ρ_{2HCi} = transverse surface relaxivity of hydrocarbon phase in pore i , m/s
 ρ_{2HCS} = transverse surface relaxivity of hydrocarbon phase in sandstone, m/s
 ρ_{2WC} = transverse surface relaxivity of water phase in carbonate, m/s
 ρ_{2Wi} = transverse surface relaxivity of water phase in pore i , m/s
 ρ_{2WS} = transverse surface relaxivity of water phase in sandstone, m/s
 ϕ_i = normalized pore-volume fraction of pore i
 ϕ_{ij} = normalized pore-volume fraction of fluid j located in pore i
 ϕ_T = total rock porosity

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Appendix A—Derivation of the New NMR-Based Wettability Index

The magnetization decay, normalized to rock porosity, obtained from NMR T_2 measurements of partially saturated mixed-wet rock samples can be written in multiexponential form as

$$M(t) = \phi_T \sum_{j=1}^N \sum_{i=1}^{\infty} \phi_{ij} e^{\frac{-t}{T_{2ij}(r)}}. \quad \text{..... (A-1)}$$

In Eq. A-1, $M(t)$ is the magnetization signal, t is time, ϕ_T is the total porosity, N is the total number of fluid components, and ϕ_{ij} and T_{2ij} are the normalized pore-volume fraction and the decay time of fluid component j in pore i , respectively (Looyestijn and Hofman 2006; Looyestijn 2008). The normalization of the magnetization signal by rock porosity values allows us to represent the fractional pore-volume and fractional pore-surface area distributions as

$$\sum_{j=1}^N \sum_{i=1}^{\infty} \phi_{ij}(r) = 1, \quad \text{..... (A-2)}$$

and

$$\sum_{j=1}^N \sum_{i=1}^{\infty} \alpha_{ij}(r) = 1, \quad \text{..... (A-3)}$$

where $\alpha_{ij}(r)$ is the normalized pore-surface area fraction of pore i normalized to the total pore surface area of the rock sample. Here, we make a distinction between the pore-volume fraction and the pore-surface area fraction distributions which was not taken into account in the publications of Looyestijn and Hofman (2006), Looyestijn (2008), and Al-Muthana et al. (2012). For pores containing both water and hydrocarbon phases, ϕ_{ij} and α_{ij} can be calculated as

$$\alpha_{ij}(r) = \alpha_i(r) W_i(r), \quad \text{..... (A-4)}$$

and

$$\phi_{ij}(r) = \phi_i(r) S_i(r), \quad \text{..... (A-5)}$$

where ϕ_i is the normalized pore-volume fraction of pore i , $\alpha_i(r)$ is the normalized pore-surface area fraction of pore i , $W_i(r)$ is the fraction of surface area of pore i in contact with water, and $S_i(r)$ is the water saturation in pore i . The overall saturation of water phase (S_w) and the overall surface area fraction covered by water (f_w) in the rock sample can be calculated as

$$\sum_{i=1}^{\infty} \phi_i(r) S_i(r) = S_w, \quad \text{..... (A-6)}$$

and

$$\sum_{i=1}^{\infty} \alpha_i(r) W_i(r) = f_w. \quad \dots\dots\dots (A-7)$$

In the fast diffusion regime, the T_2 values of the water phase in pore i can be written as

$$\frac{1}{T_{2Wi}} = \frac{1}{T_{BW}} + \frac{\rho_{2Wi} A_{Wi}}{V_{Wi}}. \quad \dots\dots\dots (A-8)$$

In Eq. A-8, T_{2Wi} is the T_2 value of water phase in pore i , ρ_{2Wi} is the transverse surface relaxivity of water phase in pore i , and A_{Wi} and V_{Wi} are the surface area and volume of water in pore i , respectively (Brownstein and Tarr 1979). Here, we assume that the influence of internal magnetic field gradients on the T_2 decay is negligible. Such an assumption is valid for the experiments with short echo-spacing times and samples with limited concentration of clay and paramagnetic components (Morris et al. 1997; Zhang et al. 2001; Tandon and Heidari 2018). Similarly, the T_2 values of the hydrocarbon phase in the pore i can be written as

$$\frac{1}{T_{2HCi}} = \frac{1}{T_{BHC}} + \frac{\rho_{2HCi} A_{HCi}}{V_{HCi}}. \quad \dots\dots\dots (A-9)$$

In Eq. A-9, T_{2HCi} is the T_2 value of hydrocarbon phase in pore i , and ρ_{2HCi} is the transverse surface relaxivity for water phase, A_{HCi} and V_{HCi} are the surface area and the volume of hydrocarbon in pore i , respectively.

The harmonic mean of the T_2 distribution for the partially saturated mixed-wet rock sample can be calculated as

$$\frac{1}{T_{2HM}} = \sum_0^{\infty} \phi_i(r) S_i(r) \frac{1}{T_{2Wi}} + \sum_0^{\infty} \phi_i(r) [1 - S_i(r)] \frac{1}{T_{2HCi}}. \quad \dots\dots\dots (A-10)$$

Eqs. A-8 and A-9 can be substituted in Eq. A-10 to obtain

$$\frac{1}{T_{2HM}} = \sum_{i=1}^{\infty} \phi_i(r) S_i(r) \left(\frac{1}{T_{BW}} + \frac{\rho_{2W} A_{Wi}}{V_{Wi}} \right) + \sum_{i=1}^{\infty} \phi_i(r) [1 - S_i(r)] \left(\frac{1}{T_{BHC}} + \frac{\rho_{2HC} A_{HCi}}{V_{HCi}} \right). \quad \dots\dots\dots (A-11)$$

Eq. A-11 can be separated into different bulk and surface components for different fluids as

$$\frac{1}{T_{2HM}} = \frac{1}{T_{BHM}} + \frac{1}{T_{2SW}} + \frac{1}{T_{2SHC}}, \quad \dots\dots\dots (A-12)$$

where

$$\frac{1}{T_{BHM}} = \sum_{i=1}^{\infty} \left(\frac{\phi_i(r) S_i(r)}{T_{BW}} + \frac{\phi_i(r) [1 - S_i(r)]}{T_{BHC}} \right), \quad \dots\dots\dots (A-13)$$

$$\frac{1}{T_{2SW}} = \sum_{i=1}^{\infty} \phi_i(r) S_i(r) \left(\frac{\rho_{2W} A_{Wi}}{V_{Wi}} \right), \quad \dots\dots\dots (A-14)$$

and

$$\frac{1}{T_{2SHC}} = \sum_0^{\infty} \phi_i(r) [1 - S_i(r)] \left(\frac{\rho_{2HC} A_{HCi}}{V_{HCi}} \right). \quad \dots\dots\dots (A-15)$$

Eq. A-13 can be simplified assuming constant values of bulk relaxivities for water and hydrocarbon phases as

$$\frac{1}{T_{BHM}} = \frac{1}{T_{2BW}} \sum_0^{\infty} \phi_i(r) S_i(r) + \frac{1}{T_{2BHC}} \sum_0^{\infty} \phi_i(r) [1 - S_i(r)]. \quad \dots\dots\dots (A-16)$$

Eq. A-16 can be simplified further using Eq. A-6 to

$$\frac{1}{T_{BHM}} = \frac{S_w}{T_{2BW}} + \frac{1 - S_w}{T_{2BHC}}. \quad \dots\dots\dots (A-17)$$

For a complex mixture of hydrocarbons present in the pore space, simplifying assumptions have to be made about the fractions and bulk decay constants of hydrocarbon components that might be located in different pores of the rock sample. If all components of the hydrocarbon mixture are assumed to exist in all the pores, then the arithmetic mean of hydrocarbon bulk distribution can be used in Eq. A-17 for calculating T_{2BHC} .

The terms A_{Wi} , and V_{Wi} can be expressed using their fundamental definitions as

$$A_{Wi} = A_P \alpha_i(r) W_i(r), \quad \dots\dots\dots (A-18)$$

and

$$V_{Wi} = V_P \phi_i(r) S_i(r), \quad \dots\dots\dots (A-19)$$

where V_P and A_P are the total PV and the total pore area of the rock sample, respectively. Eqs. A-18 and A-19 can be substituted in Eqs. A-14 and A-15, respectively, to yield

$$\frac{1}{T_{2SW}} = \sum_0^{\infty} \phi_i(r) S_i(r) \left(\frac{\rho_{2W} A_P \alpha_i(r) W_i(r)}{V_P \phi_i(r) S_i(r)} \right), \dots\dots\dots (A-20)$$

and

$$\frac{1}{T_{2SHC}} = \sum_0^{\infty} \phi_i(r) [1 - S_i(r)] \left(\frac{\rho_{2HC} A_P \alpha_i(r) [1 - W_i(r)]}{V_P \phi_i(r) [1 - S_i(r)]} \right). \dots\dots\dots (A-21)$$

Eqs. A-20 and A-21 can be further simplified assuming constant average values of ρ_{2HC} and ρ_{2W} as

$$\frac{1}{T_{2SW}} = \rho_{2W} \frac{A_P}{V_P} \sum_0^{\infty} \alpha_i(r) W_i(r), \dots\dots\dots (A-22)$$

and

$$\frac{1}{T_{2SHC}} = \rho_{2HC} \frac{A_P}{V_P} \sum_0^{\infty} \alpha_i(r) [1 - W_i(r)]. \dots\dots\dots (A-23)$$

The summation terms in Eqs. A-22 and A-23 can be simplified using Eq. A-7 to

$$\frac{1}{T_{2SW}} = \rho_{2W} \frac{A_P}{V_P} f_W, \dots\dots\dots (A-24)$$

and

$$\frac{1}{T_{2SHC}} = \rho_{2HC} \frac{A_P}{V_P} (1 - f_W). \dots\dots\dots (A-25)$$

Combining Eqs. A-17, A-24, and A-25, Eq. A-12 can be expressed as

$$\frac{1}{T_{2HM}} = \frac{1}{T_{BHM}} + \rho_{2W} \frac{A_P}{V_P} f_W + \rho_{2HC} \frac{A_P}{V_P} (1 - f_W). \dots\dots\dots (A-26)$$

Rearranging Eq. A-26, the value of f_W can be written as

$$f_W = \frac{\left(\frac{1}{T_{2HM}} - \frac{1}{T_{BHM}} \right) - \rho_{2HC} \frac{A_P}{V_P}}{\rho_{2W} \frac{A_P}{V_P} - \rho_{2HC} \frac{A_P}{V_P}}. \dots\dots\dots (A-27)$$

The new NMR-based wettability index (I_{NMR}) can then be computed from the value of f_W using Eq. 6 as

$$I_{NMR} = 2f_W - 1. \dots\dots\dots (A-28)$$

The value of f_W can be substituted in Eq. A-28 from Eq. A-27 to obtain the value of I_{NMR} as

$$I_{NMR} = \frac{2 \left(\frac{1}{T_{2HM}} - \frac{1}{T_{BHM}} \right) - \left(\frac{\rho_{2W} A_P}{V_P} + \frac{\rho_{2HC} A_P}{V_P} \right)}{\frac{\rho_{2W} A_P}{V_P} - \frac{\rho_{2HC} A_P}{V_P}}. \dots\dots\dots (A-29)$$

The values of surface relaxivities and overall surface-to-volume ratios of the rock sample can be substituted using the harmonic mean of the NMR responses for fully water-saturated water-wet and fully hydrocarbon-saturated hydrocarbon-wet rock samples of the same rock type as

$$\frac{1}{T_{2WHM}} = \sum_0^{\infty} \phi_i(r) \left[\frac{1}{T_{BW}} + \frac{\rho_{2W} A_P \alpha_i(r)}{V_P \phi_i(r)} \right], \dots\dots\dots (A-30)$$

and

$$\frac{1}{T_{2HCHM}} = \sum_0^{\infty} \phi_i(r) \left[\frac{1}{T_{BHC}} + \frac{\rho_{2HC} A_P \alpha_i(r)}{V_P \phi_i(r)} \right]. \dots\dots\dots (A-31)$$

Eqs. A-30 and A-31 can be substituted in Eq. A-29 to yield the expression for I_{NMR} as

$$I_{NMR} = \frac{2 \left(\frac{1}{T_{2HM}} - \frac{1}{T_{BHM}} \right) - \left[\left(\frac{1}{T_{2WHM}} - \frac{1}{T_{BW}} \right) + \left(\frac{1}{T_{2HCHM}} - \frac{1}{T_{BHC}} \right) \right]}{\left(\frac{1}{T_{2WHM}} - \frac{1}{T_{BW}} \right) - \left(\frac{1}{T_{2HCHM}} - \frac{1}{T_{BHC}} \right)} \dots\dots\dots (A-32)$$

Appendix B—Simulated NMR T_2 Distributions in all the Pore-Scale Images Used in this Study

Fig. B-1 shows the simulated T_2 distributions in all the pore-scale samples used in this study.

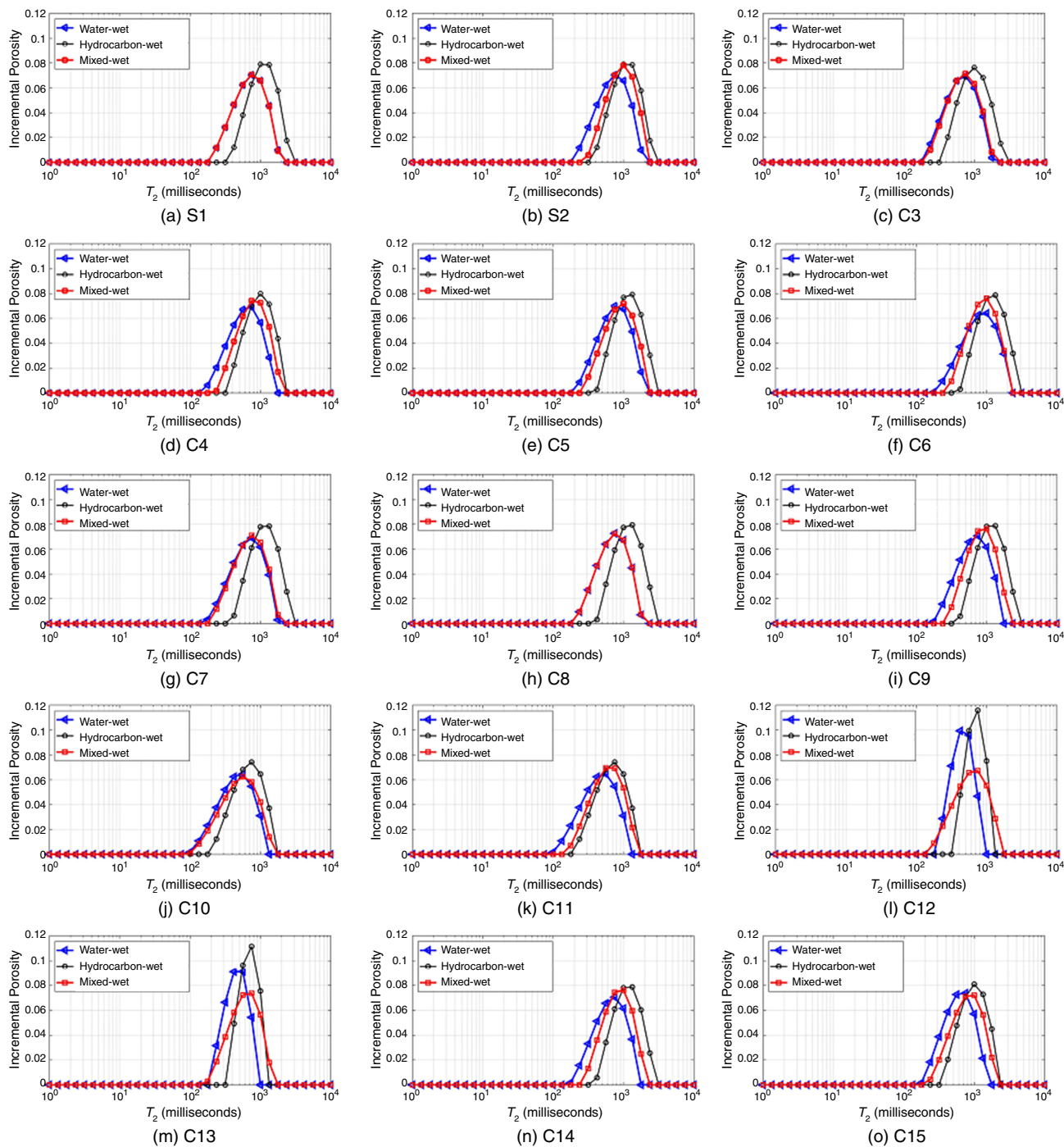


Fig. B-1—Simulated T_2 distribution, showing the influence of wettability and fluid saturation on NMR response in rock samples (a) S1, (b) S2, (c) C3, (d) C4, (e) C5, (f) C6, (g) C7, (h) C8, (i) C9, (j) C10, (k) C11, (l) C12, (m) C13, (n) C14, and (o) C15. The blue, black, and red curves show the T_2 distribution for fully water-saturated water-wet, fully hydrocarbon-saturated hydrocarbon-wet, and partially saturated mixed-wet rock samples, respectively.

Appendix C—Water Saturations and NMR T_2 Peaks for the TC Core Samples

Table C-1 lists the measured water saturation values and the corresponding water and hydrocarbon T_2 distribution peaks for each core sample given in Fig. 8.

Sample	I_{AH}	S_W (v/v)	T_{2WP} (milliseconds)	T_{2HCP} (milliseconds)	T_{2HM} (milliseconds)	T_{BHM} (milliseconds)
TC-A	0.5	0.38	34	1080	29	1735
		0.40	31	909	29	1761
		0.43	34	1000	28	1788
		0.44	38	1000	25	1790
		0.47	38	1000	26	1827
TC-B	0.2	0.42	38	909	34	1773
		0.43	38	1000	36	1784
		0.47	38	1000	34	1834
		0.49	38	1100	32	1847
		0.52	38	1100	33	1880
TC-C	0.1	0.50	38	909	32	1861
		0.52	38	909	33	1882
		0.57	38	1000	32	1939
		0.59	38	1100	29	1953
		0.61	38	1100	29	1980
TC-D	-0.6	0.27	56	909	78	1625
		0.29	56	909	82	1658
		0.35	56	826	75	1711
		0.41	56	826	72	1767
		0.50	56	681	81	1858

Table C-1—Measured water saturations, T_2 peaks, and harmonic mean of T_2 distribution for all the core samples.